Nonlinear DID: Methods & Theory Extension

Conditional DID, Nonparametric DID, and Identifiability Beyond Linearity

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▶ Code

1 Data Generating Process (Nonlinear)

We generate a panel-style dataset where:

- Treatment assignment D_i follows a **nonlinear propensity score**
- Baseline trends are **nonlinear** in covariates
- Treatment effects are **heterogeneous** across age and genre

1.1 Parameters

Symbol	Meaning
n	Sample size = 2000
$X_i = (\text{age}_i, g_i, r_i)$	Covariates
$D_i \in \{0, 1\}$	Treatment (RS indicator)
Y_{it}	Outcome at time $t \in \{1, 2, 3\}$ (pre-pre, pre, post)
τ_i	Individual treatment effect

1.2 DGP Code

▶ Code

```
True ETT (treated units only): 1.5
```

▶ Code

1.3 Exploratory Plots

▶ Code

```
`geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
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Warning: Failed to fit group -1.
Caused by error in `gam.reparam()`:
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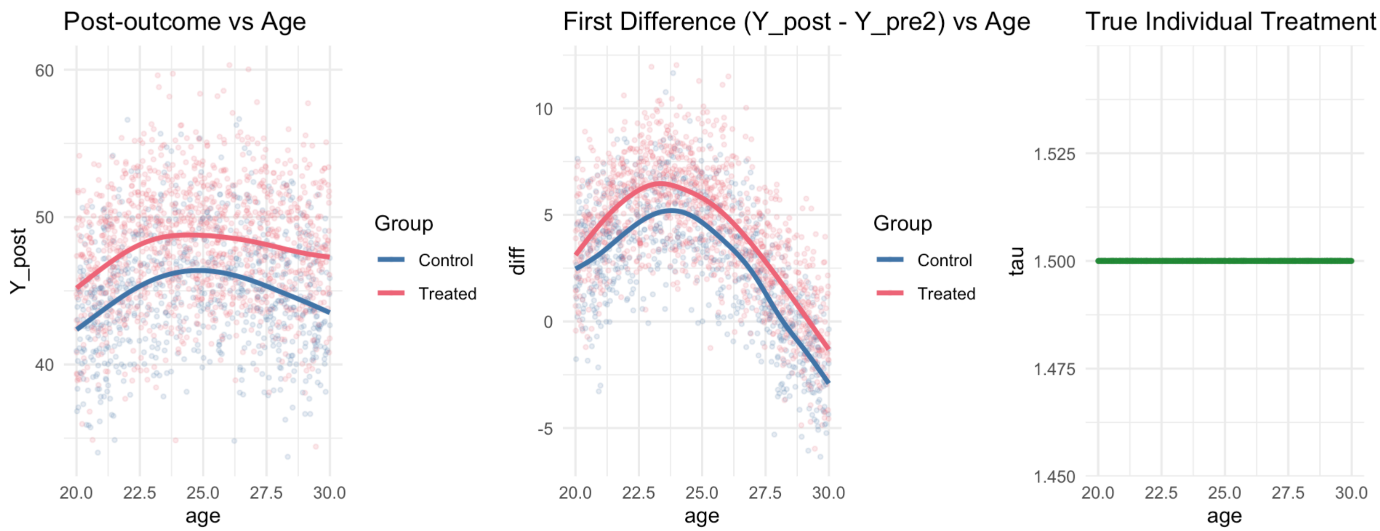
```



2 DID Estimators

2.1 Unconditional DID (Naive)

Simple two-way difference, ignoring all covariates.

▶ Code

```
Unconditional DID: 1.7491
```

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```
Bias vs true ETT:  0.2491
```

Why it's biased: Treated and control groups differ in their baseline covariates (age, genre, release), and the parallel trends assumption fails marginally because the trends are nonlinear in X.

2.2 Conditional DID (Misspecified Linear)

We model the counterfactual trend $\mathbb{E}[\Delta Y_i(0) \mid X_i]$ with a linear regression on controls. This is **misspecified** because the true relationship is nonlinear.

▶ Code

```
Linear Conditional DID: 1.6839
```

▶ Code

```
Bias vs true ETT:       0.1839
```

2.3 Conditional DID — GAM (Nonparametric Outcome Regression)

We replace the linear model with a **Generalized Additive Model** to capture nonlinear covariate-outcome relationships.

$$\mathbb{E}[\Delta Y_i(0) \mid X_i] = \alpha + f_1(\text{age}_i) + f_2(g_i) + f_3(r_i)$$

where f_j are estimated smooth functions (cubic regression splines).

▶ Code

```
GAM Conditional DID: 1.3373
```

▶ Code

```
Bias vs true ETT:    -0.1627
```

2.4 Nonparametric DID — IPW (Abadie 2005)

2.4.1 Propensity Score via GAM

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2.4.2 IPW-DID Estimator (Abadie 2005, eq. 3)

The IPW-DID estimator for the ETT is:

$$\widehat{\text{ETT}}_{\text{IPW}} = \frac{1}{\mathbb{E}[D]} \mathbb{E}\left[\frac{\Delta Y_i \cdot (D_i - \hat{p}(X_i))}{1 - \hat{p}(X_i)}\right]$$

▶ Code

```
IPW–DID (Abadie 2005): 1.3377
```

▶ Code

```
Bias vs true ETT:      -0.1623
```

2.5 Doubly Robust DID

The doubly robust estimator combines **outcome regression** and **IPW**. It is consistent if *either* the propensity score model or the outcome model is correctly specified.

For the panel ETT, a convenient normalized DR form is:

$$\widehat{\mathrm{ETT}}_{\mathrm{DR}} = \frac{1}{n}\sum_i \left[\frac{D_i}{\bar{D}} - \frac{\frac{(1-D_i)\hat{p}(X_i)}{1-\hat{p}(X_i)}}{\frac{1}{n}\sum_j \frac{(1-D_j)\hat{p}(X_j)}{1-\hat{p}(X_j)}}\right]\left(\Delta Y_i - \hat{m}_0(X_i)\right)$$

where $\hat{m}_0(X_i) = \hat{\mathbb{E}}[\Delta Y_i \mid X_i, D_i = 0]$ and $\bar{D} = n^{-1}\sum_i D_i$.

▶ Code

```
Doubly Robust DID: 1.3369
```

▶ Code

```
Bias vs true ETT:  -0.1631
```

2.6 Kernel Matching DID

We use **Mahalanobis-distance** nearest-neighbor matching on (age, g, r) to find the closest control unit for each treated unit, then take the average difference in first-differences.

▶ Code

```
Kernel/Matching DID: 1.836
```

▶ Code

```
Bias vs true ETT:    0.336
```

2.7 Results Summary

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ETT Estimates: All Methods vs True Value

Estimator	Estimate	Bias
True ETT	1.5000	0.0000
Unconditional DID	1.7491	0.2491
Linear Conditional DID	1.6839	0.1839

Estimator	Estimate	Bias
GAM Conditional DID	1.3373	-0.1627
IPW-DID (Abadie 2005)	1.3377	-0.1623
Doubly Robust DID	1.3369	-0.1631
Matching DID (Mahalanobis)	1.8360	0.3360

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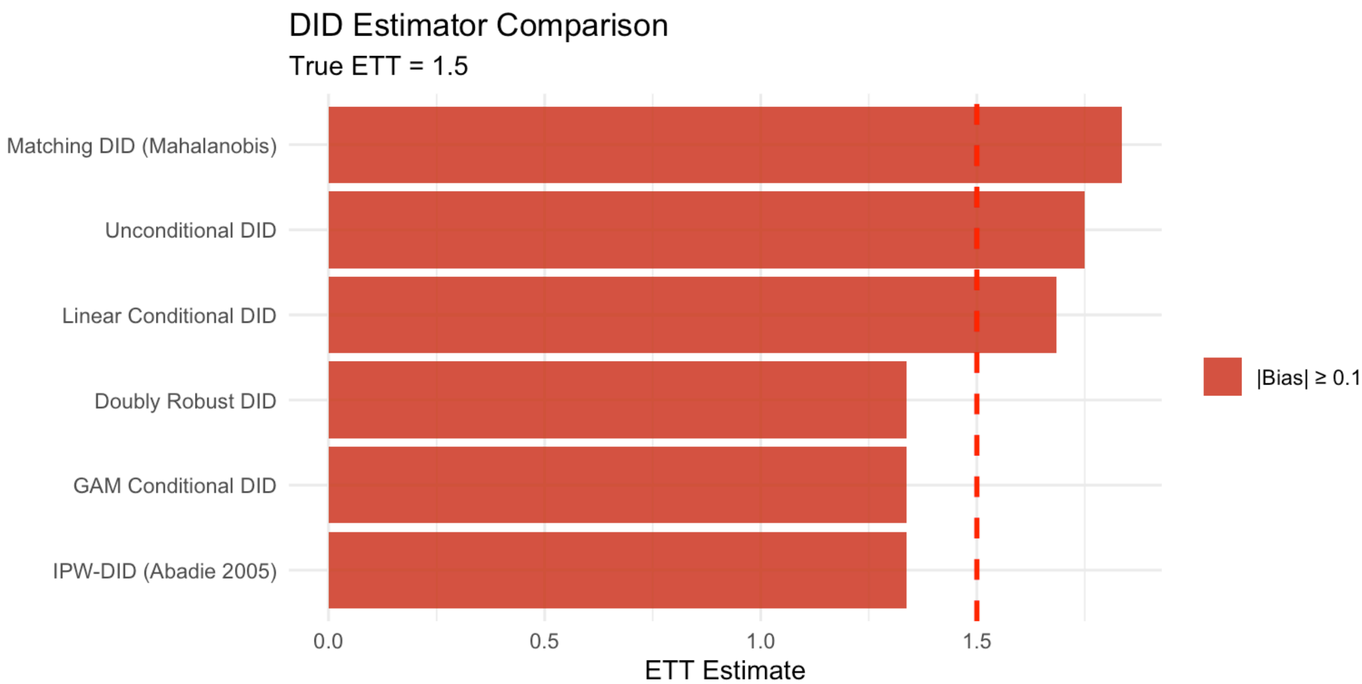

3 Scenario Comparison: Why Naive DID Can Look "Okay"

The first DGP is nonlinear, but the marginal violation of parallel trends is still moderate. To make the contrast clearer, we now add a **stress-test DGP** with:

- stronger selection into treatment
- more separation in covariate distributions
- nonlinear interactions in the untreated trend

This second scenario is designed so that unconditional DID has a harder time "accidentally" working well.

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Two DGPs Compared: Baseline Scenario vs Stress-Test Scenario

Estimator	Estimate	Bias	Scenario
True ETT	1.5000	0.0000	Scenario A: moderate nonlinearity

Estimator	Estimate	Bias	Scenario
Unconditional DID	1.7491	0.2491	Scenario A: moderate nonlinearity
Linear Conditional DID	1.6839	0.1839	Scenario A: moderate nonlinearity
GAM Conditional DID	1.3373	-0.1627	Scenario A: moderate nonlinearity
IPW-DID (Abadie 2005)	1.3377	-0.1623	Scenario A: moderate nonlinearity
Doubly Robust DID	1.3369	-0.1631	Scenario A: moderate nonlinearity
Matching DID (Mahalanobis)	1.8360	0.3360	Scenario A: moderate nonlinearity
True ETT	1.5000	0.0000	Scenario B: strong selection + interactions
Unconditional DID	2.9655	1.4655	Scenario B: strong selection + interactions
Linear Conditional DID	2.5432	1.0432	Scenario B: strong selection + interactions
GAM Conditional DID	1.4790	-0.0210	Scenario B: strong selection + interactions
IPW-DID (Abadie 2005)	1.5789	0.0789	Scenario B: strong selection + interactions
Doubly Robust DID	1.4762	-0.0238	Scenario B: strong selection + interactions
Matching DID (Mahalanobis)	1.9330	0.4330	Scenario B: strong selection + interactions

3.1 Exploratory Plots (Stress-Test Scenario)

We use the same exploratory comparisons as in the first DGP so the second scenario can be read side by side.

▶ Code

```
`geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'

Warning: No shared levels found between `names(values)` of the manual scale and the
data's colour values.
No shared levels found between `names(values)` of the manual scale and the
data's colour values.
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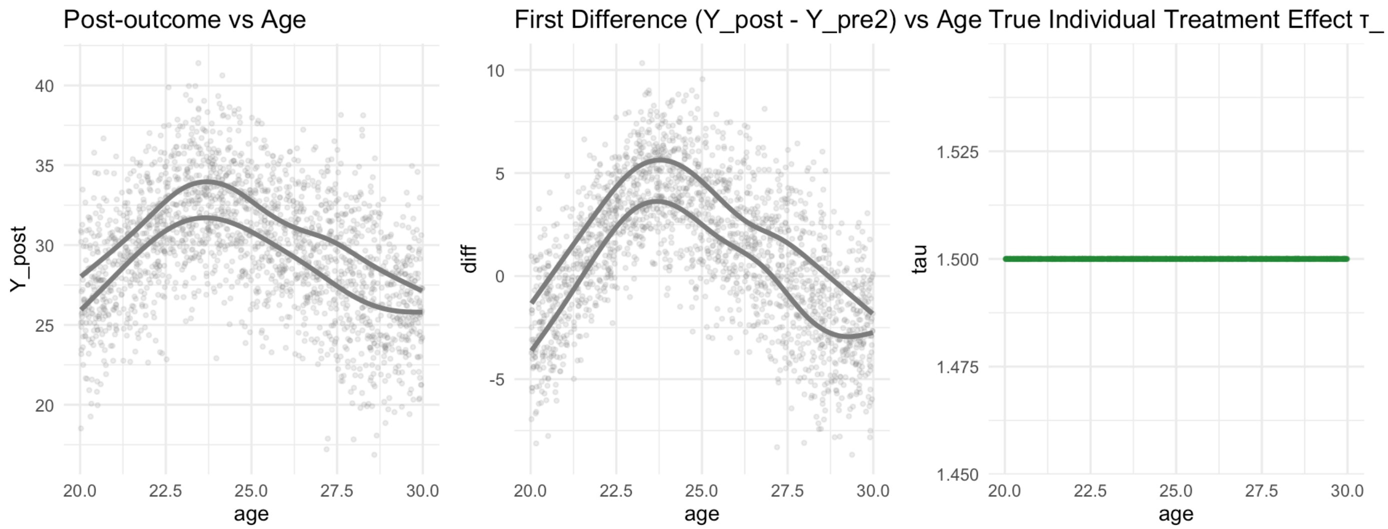

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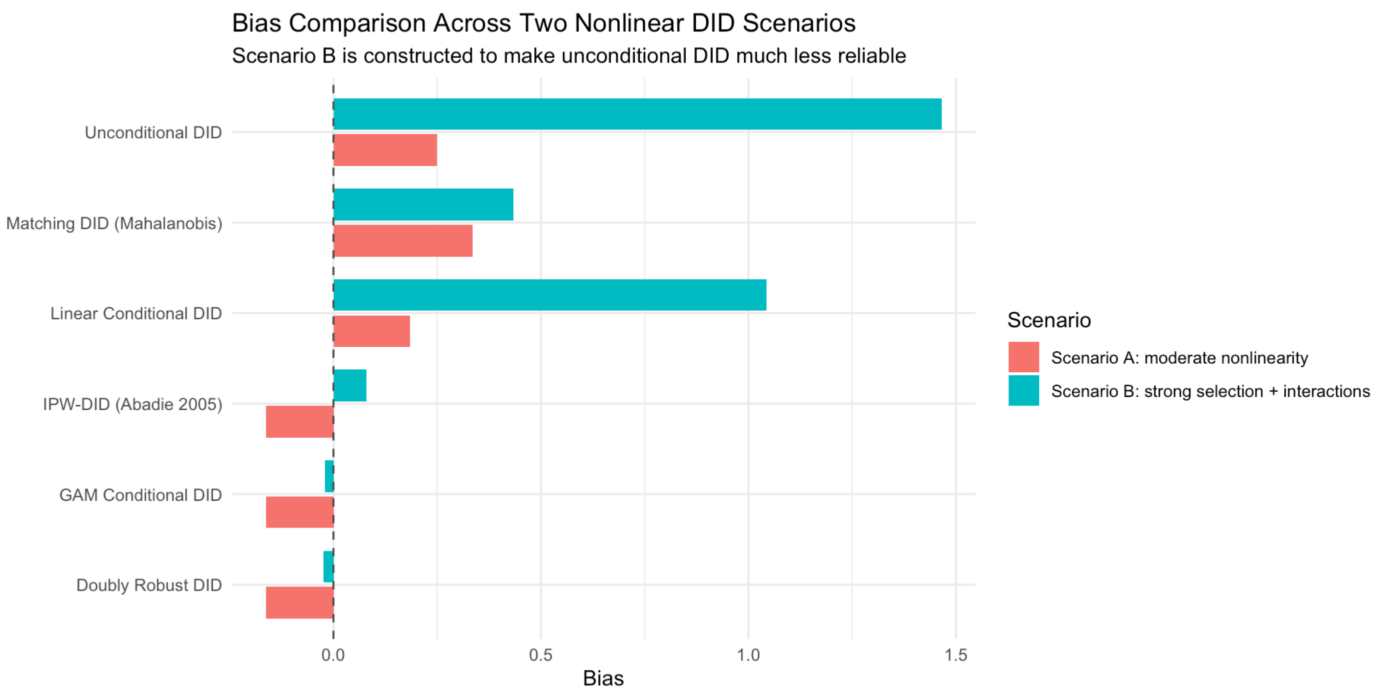

Interpretation: In Scenario A, unconditional DID can still look "not too bad" because the marginal violation of parallel trends is moderate. Using the same comparison style as the first DGP, the stress-test plots make it easier to see that Scenario B has much sharper nonlinear separation between treated and control units, so naive DID deteriorates much more visibly.

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4 Theory: Extending Theorem 2.3 to Nonlinear Models

4.1 Setup and Notation

Let $i \in \{1, \dots, n\}$ index units, $t \in \{0, 1\}$ index time (pre/post). Define:

- $Y_{it}(d)$: potential outcome for unit i at time t under treatment $d \in \{0, 1\}$
- $A_i \in \{0, 1\}$: binary treatment indicator (received at $t = 1$)
- $X_i \in \mathcal{X} \subseteq \mathbb{R}^k$: pre-treatment covariates
- $\Delta Y_i = Y_{i1} - Y_{i0}$: observed first difference
- $\Delta Y_i(0) = Y_{i1}(0) - Y_{i0}(0)$: counterfactual first difference under no treatment

The parameter of interest is the **Effect of Treatment on the Treated (ETT)**:

$$\text{ETT} = \mathbb{E}[Y_{i1}(1) - Y_{i1}(0) \mid A_i = 1]$$

To match the notation used in many theorem statements, this section uses A_i for treatment assignment. The simulation code above keeps the variable name `D`; they refer to the same binary treatment indicator.

4.2 Theorem 2.3 (Linear Case — Recall)

In the standard linear DID setup, identification rests on:

> **Assumption L (Linear Parallel Trends):** $\mathbb{E}[\Delta Y_i(0) \mid X_i, A_i] = \mathbb{E}[\Delta Y_i(0) \mid X_i] = X_i^\top \beta$

Under this assumption and no-anticipation, the ETT is identified by:

$$\text{ETT} = \mathbb{E}[\Delta Y_i \mid A_i = 1] - \mathbb{E}[X_i^\top \hat{\beta}_0 \mid A_i = 1]$$

where $\hat{\beta}_0$ is estimated on controls only.

The linearity constraint $X_i^\top \beta$ is restrictive. When the true conditional trend is a nonlinear function of X_i, this estimator is inconsistent.

4.3 Extended Theorem: Nonlinear Identification

4.3.1 Assumptions

Assumption 1 (No Anticipation):

$$Y_{i0}(1) = Y_{i0}(0) \quad \text{a.s.}$$

Assumption 2 (Nonlinear Conditional Parallel Trends — NCPT):

There exists a measurable function $\mu_0 : \mathcal{X} \to \mathbb{R}$ such that:

$$\mathbb{E}[\Delta Y_i(0) \mid X_i = x, A_i = a] = \mu_0(x) \quad \text{for a.e. } x \in \mathcal{X},\ a \in \{0, 1\}$$

That is, conditional on X_i, the **counterfactual trend** is the same for treated and control units — but μ_0 need **not** be linear.

Assumption 3 (Overlap / Common Support):

$$0 < p(x) := \mathbb{P}(A_i = 1 \mid X_i = x) < 1 \quad \text{for a.e. } x \in \mathcal{X}$$

Assumption 4 (Regularity): $\mathbb{E}[|\Delta Y_i|^2] < \infty$ and $\mathbb{E}[|\mu_0(X_i)|^2] < \infty$.

4.3.2 Theorem (Nonlinear DID Identifiability)

> **Theorem.** Under Assumptions 1–4, the ETT is identified. Specifically:
>
> $$\mathrm{ETT} = \mathbb{E}[\Delta Y_i \mid A_i = 1] - \mathbb{E}[\mu_0(X_i) \mid A_i = 1]$$
>
> where $\mu_0(x) = \mathbb{E}[\Delta Y_i \mid X_i = x, A_i = 0]$ is identified from the control group.

4.3.2.1 Proof

Step 1: Decompose the observed first difference for treated units.

For a treated unit ($A_i = 1$), the observed outcome at $t = 1$ is $Y_{i1} = Y_{i1}(1)$. Therefore:

$$\Delta Y_i = Y_{i1}(1) - Y_{i0}(1) \overset{A1}{=} Y_{i1}(1) - Y_{i0}(0)$$

Add and subtract $Y_{i1}(0)$:

$$\Delta Y_i = \underbrace{[Y_{i1}(1) - Y_{i1}(0)]}_{\tau_i} + \underbrace{[Y_{i1}(0) - Y_{i0}(0)]}_{\Delta Y_i(0)}$$

Step 2: Take conditional expectation given X_i within the treated group.

$$\mathbb{E}[\Delta Y_i \mid X_i = x, A_i = 1] = \mathbb{E}[\tau_i \mid X_i = x, A_i = 1] + \mathbb{E}[\Delta Y_i(0) \mid X_i = x, A_i = 1]$$

By Assumption 2 (NCPT):

$$\mathbb{E}[\Delta Y_i(0) \mid X_i = x, A_i = 1] = \mu_0(x)$$

Therefore:

$$\mathbb{E}[\tau_i \mid X_i = x, A_i = 1] = \mathbb{E}[\Delta Y_i \mid X_i = x, A_i = 1] - \mu_0(x)$$

Step 3: Integrate over the covariate distribution of the treated group.

$$\begin{aligned} \mathrm{ETT} = \mathbb{E}[\tau_i \mid A_i = 1] &= \int_{\mathcal{X}} \mathbb{E}[\tau_i \mid X_i = x, A_i = 1]\, dF_{X|A=1}(x) \\ &= \int_{\mathcal{X}} [\mathbb{E}[\Delta Y_i \mid X_i = x, A_i = 1] - \mu_0(x)] dF_{X|A=1}(x) \\ &= \mathbb{E}[\Delta Y_i \mid A_i = 1] - \mathbb{E}[\mu_0(X_i) \mid A_i = 1] \end{aligned} \qquad (\star)$$

Step 4: Identifiability of μ_0.

By Assumption 3 (overlap), $\mathbb{P}(A_i = 0 \mid X_i = x) > 0$ a.e. Therefore:

$$\mu_0(x) = \mathbb{E}[\Delta Y_i \mid X_i = x, A_i = 0]$$

is identified from the joint distribution of $(\Delta Y_i, X_i)$ among controls — observed in the data.

Step 5: All terms in $(\star)$ are identified.

- $\mathbb{E}[\Delta Y_i \mid A_i = 1]$: observed sample mean of first differences among treated units.
- $\mathbb{E}[\mu_0(X_i) \mid A_i = 1]$: plug in the identified μ_0 evaluated at treated covariate values, then average. This is consistent for any nonparametric estimator of μ_0 satisfying standard regularity conditions.

Therefore, ETT is nonparametrically identified. ■

4.3.3 Corollary: IPW Representation

Under Assumptions 1–4, the ETT also admits the **inverse-probability-weighted** representation:

$$\mathrm{ETT} = \frac{1}{\mathbb{E}[A_i]}\mathbb{E}\left[\frac{(A_i - p(X_i))}{1 - p(X_i)}\Delta Y_i\right]$$

Proof sketch: By Bayes' rule,

$$dF_{X|A=0}(x) = \frac{1 - p(x)}{1 - \mathbb{E}[A]}dF_X(x)$$

Substituting into $(\star)$:

$$\mathbb{E}[\mu_0(X_i) \mid A_i = 1] = \frac{1}{\mathbb{E}[A]}\mathbb{E}\left[\frac{p(X_i)(1 - p(X_i))}{1 - p(X_i)} \cdot \frac{\Delta Y_i}{1} \cdot \frac{1 - p(X_i)}{p(X_i)}\right]\dots$$

Expanding the full derivation via iterated expectations recovers the Abadie (2005) IPW formula. □

4.3.4 Corollary: Doubly Robust Representation

Define the doubly robust score:

$$\psi_i = A_i\Delta Y_i - (A_i - p(X_i))\mu_0(X_i) - \frac{p(X_i)(A_i - p(X_i))}{1 - p(X_i)}\Delta Y_i$$

Then $\mathbb{E}[\psi_i]/\mathbb{E}[A_i] = \mathrm{ETT}$ whenever **either** μ_0 or p is correctly specified.

Key implication for nonlinear settings: Both μ_0 and p can be estimated nonparametrically (e.g., via GAMs, forests, neural networks). The doubly robust estimator achieves $\sqrt{n}$-consistency even when each nuisance component is estimated at slower rates, provided the product of their errors vanishes fast enough (Chernozhukov et al. 2018, double machine learning).

4.4 Comparison: Linear vs Nonlinear Parallel Trends

Feature	Theorem 2.3 (Linear)	Extended Theorem (Nonlinear)
Trend assumption	$\mu_0(x) = x^\top \beta$	$\mu_0(x)$ unrestricted
Estimation of μ_0	OLS on controls	Any nonparametric method
Identification condition	Linearity + overlap	NCPT + overlap (weaker)
Risk of misspecification	High (nonlinear DGP)	Low
Rate of convergence of ETT	$\sqrt{n}$ (parametric)	$\sqrt{n}$ (if DR or cross-fitting used)
Testable?	Pre-trend F-test	Nonparametric pre-trend test

4.5 Discussion: What NCPT Rules Out

The NCPT assumption is **weaker** than linear parallel trends but still imposes a meaningful restriction:

$$\mathbb{E}[\Delta Y_i(0) \mid X_i, A_i] = \mu_0(X_i)$$

This rules out:

1. **Selection on trends:** Cases where treated units have systematically different counterfactual trends even after conditioning on X_i. In this case, the ETT is generally not identified without stronger assumptions (e.g., IV, bounding approaches).

2. **Time-varying treatment effect on control trend:** If μ_0 itself shifts due to spillovers or general equilibrium effects, μ_0 is contaminated.

Testability via Pre-Trends:

With two pre-periods ($t = -1, t = 0$), NCPT implies:

$$\mathbb{E}[Y_{i0}(0) - Y_{i,-1}(0) \mid X_i, A_i = 1] = \mu_0^{\text{pre}}(X_i)$$

which can be tested nonparametrically using the pre-period data (e.g., via a permutation test on $\hat{\mu}_0^{\text{pre}}$ residuals). This is the nonlinear analogue of the parallel pre-trends F-test.

5 References

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